

Beyond prediction: AI agents in the next era of environmental early warning

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When an artificial intelligence (AI)-driven weather model signals catastrophic wildfire risk days in advance, but evacuation orders still arrive too late, whose failure is it? This is no longer a hypothetical. In January 2025, the Los Angeles wildfires burned far more area than Southern California Edison's internal AI forecasts had projected (<https://www.reuters.com/business/environment/los-angeles-wildfires-were-10-times-bigger-than-utilities-ai-forecast-2025-06-09/>). Weeks later, *Reuters* reported that a federal online disaster planning tool faced shutdown amid institutional budget cuts (<https://www.reuters.com/business/environment/us-online-disaster-planning-tool-may-go-dark-wednesday-agency-website-says-2025-09-09/>). The *Washington Post* report on Hurricane Helene's inland flooding in North Carolina similarly suggested that a new era of extreme flood events is exceeding the coordination capacity of the agencies responsible for emergency response (<https://www.washingtonpost.com/climate-environment/2025/07/07/hurricane-helene-evacuation-north-carolina-warnings/>). Together, these cases point to a deeper dilemma: AI has revolutionized disaster detection and forecasting, but translating accurate forecasts into effective environmental emergency response remains unresolved (Wang & Lu, 2026).

The coordination gap and the rise of AI agents

Artificial intelligence has revolutionized the forecast of extreme environmental events, including floods, droughts, wildfires, and heatwaves (Camps-Valls et al., 2025). Google's GraphCast generates a 10-day global forecast in under a minute (Price et al., 2025). Yet predictive accuracy does not guarantee effective response. Wang and Lu (2026) argue that most high-accuracy predictions still fail to trigger preventive actions. This is not because the forecasts are insufficiently accurate, but because AI-driven prediction models often create new data silos rather than supportive environments for shared decision-making.

If forecasts lack transparency, emergency managers are understandably reluctant to rely on them under time pressure (Musse, 2026). Effective emergency management depends on close

coordination among meteorological agencies, public health departments, infrastructure operators, and local governments, but this is precisely what current AI-driven models do not support. Kuglitsch et al. (2024) show that the main failures in AI-assisted early warning are institutional rather than technical. Communication channels are fragmented, authority is often unclear, and trust between forecast producers and decision-makers remains weak. The World Meteorological Organization's Early Warnings for All initiative reflects the same view, treating effective warning as a governance challenge spanning the full chain from risk knowledge and forecasting to communication and response, rather than as a forecasting problem alone (<https://wmo.int/site/knowledge-hub/programmes-and-initiatives/early-warnings-all-ew4all>). This is not simply a data problem. It is a coordination problem.

In response to this gap, multi-agent systems (MAS) are beginning to attract attention. Artificial intelligence agents are autonomous software systems capable of processing inputs, reasoning over them, and recommending or supporting action. In operational settings, these capabilities are likely to be deployed not through a single agent but through coordinated MAS. In particular, they can operate across the fragmented institutional landscape of emergency response by integrating satellite imagery and sensor feeds, routing alerts to relevant agencies, identifying inter-jurisdictional conflicts, and escalating decisions to human commanders. Chen et al. (2026) demonstrate the near-term feasibility of this approach, showing that large vision language models can generate structured postdisaster damage assessments from aerial imagery at speeds and scales beyond manual capacity. The broader implication is that environmental emergency response is moving toward linked AI functions rather than isolated predictive tools (Yanites et al., 2025). Once predisaster prediction, real-time damage assessment, and interagency response support are connected, an MAS can operate as coordination infrastructure that links information, priorities, and actions across institutions. However, the same autonomy that makes MAS attractive for emergency coordination also introduces new risks. Without sufficient validation and supervision, agent-based systems may generate false alarms, trigger unnecessary evacuations, or fail to escalate genuine warnings in time, producing unintended harms at scale.

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Governing AI agents as coordination infrastructure

These risks suggest a different regulatory and managerial framing. Artificial intelligence agents in environmental emergencies should be governed as coordination infrastructure, not merely as enhanced forecasting tools. That framing carries three practical implications. First, evaluation frameworks should extend beyond predictive accuracy to include coordination outcomes such as warning to response time, interoperability across agencies, restoration speed, and whether warnings actually improved protective behavior. This is consistent with the World Meteorological Organization's position that effective early warning systems must span the full chain from risk knowledge to response functions in practice (<https://wmo.int/site/knowledge-hub/programmes-and-initiatives/early-warnings-all-ew4all>). Predictive skill scores alone cannot reveal whether the last mile was closed.

Second, agencies should require structured validation and human authorization before deploying MAS in operational emergency settings. Validation should include stress-testing against false alarm scenarios; assessment of failure modes under data degradation; and review of potential misuse risks, particularly in systems where agent outputs could trigger mass public alerts. Once deployed, every agent-generated recommendation or action should be classified in advance as informational, review-required, or escalation-required, so that the decision-making process remains under human supervision. This would clarify what may be automated, what still requires human approval, and what must remain under incident command. Although AI agents handle microlevel hazard detection and immediate system balancing, humans must retain oversight to interpret the "black box" and enforce ethical resource allocation (Mussi et al., 2025). Tiggeloven et al. (2025) specifically warn that AI early warning tools should not be deployed without robust guardrails and institutional validation frameworks regardless of their predictive performance in controlled settings.

Third, regulators and emergency managers should mandate auditable agent-to-agent logs, including role descriptions, hand-off records, escalation rules, and postevent review trails. If multi-agent coordination is to shape real environmental emergency response, its internal decision-making process must be reconstructable after the fact. Auditability serves two purposes that prediction metrics cannot. It allows institutions to learn from coordination failures, and it gives the public a basis for judging whether these systems are being used responsibly. Without such transparency, communities cannot tell whether a warning delay resulted from data error, flawed interagent logic, human override, or organizational bottlenecks, and they have little basis for demanding accountability from any of them.

Conclusion

Artificial intelligence agents may indeed help close the last mile gap in environmental emergency response. But the decisive question is not whether they can offer more recommendations faster. It is whether they can help emergency management institutions act together transparently, equitably, and in time. If prediction defined the challenge of the last wave of early warning

systems, MAS coordination may be the defining dilemma of the next.

Data availability

No original data were used in this study. All sources cited are publicly available via the references.

Author contributions

Bo Shao (Conceptualization, Writing—original draft) and Ling Zhang (Supervision, Validation, Writing—review & editing)

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Conflicts of interest

This manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. We declare that we have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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